Cascaded Adaptation Framework for Fast Calibration of Myoelectric Control

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***Abstract*— In spite of several decades of intensive research and development, the existing algorithms of myoelectric pattern recognition (MPR) are yet to make significant clinical and commercial impact. This study focuses on the one of the limiting factors of current algorithms: degradation of algorithm performance due to the inherent non-stationarity in electromyography (EMG) signals and the consequent need for frequent re-training and re-calibration. In order to reduce the re-calibration time required for donning/doffing between sessions and avoid to need to re-calibrate within a given session while donning, we propose a cascaded adaptation (CA) framework based on linear discriminant analysis (LDA), which automatically incorporates models from previous sessions in the model calibration for the current session. The framework also updates the model parameters according to new data samples and the corresponding recognized labels. Both off-line analysis (with data from eight intact-limbed subjects and three trans-radial amputees) and online testing with 9 intact-limbed subjects were conducted to evaluate the proposed method. Results show that the LDA embedded with CA (LDA-CA) is able to classify 11 types of motion with a small training data set, beginning from the second session of the experiment. The proposed LDA-CA obtains better performance as compared with three other methods—baseline LDA (LDA-BL), LDA with self-enhancing (LDA-SE), and LDA with domain adaptation (LDA-DA). The online test demonstrates that LDA-CA requiring an initial 1 min training session can be reliably used for 8 h without re-training. The proposed myoelectric control framework with low calibration burden has the potential to move the MPR based prostheses from academic research to clinical application.**

***Index Terms*— Adaptation, electromyography, linear discriminant analysis, pattern recognition, upper extremity prostheses.**

I. Introduction

ALTHOUGH significantly intensive research efforts have been invested in myoelectric pattern recognition (MPR) for controlling upper extremity prostheses over the past two decades, few control systems have been implemented in clinical/commercial settings [1]. One of the key factors of such lack of practical impact of MPR is that experimental conditions in academic research are too simple and ideal for practical conditions in activities of daily living (ADL), including, but not limited to the following: short versus long experiment time, fixed versus changing limb position, fixed versus changing electrode positions and off-line versus real time application. Previous studies have demonstrated that many factors, such as the change of limb position, shift of electrodes and force variation, can significantly deteriorate the MPR performance [2]–[4]. In order to bridge the gap between academic research and corresponding commercial systems, the focus of recent research has seen a shift from the pure off-line classification accuracy in an ideal laboratory setting to enhancing the system robustness against these factors in ADL [2]–[9]. In addition, the real-time usability and controllability has been increasingly emphasized over classification accuracy when it comes to performance benchmarking [10]–[15].

In this study, we focus on developing an adaptive framework that is capable of robustness against non-stationarity of the electromyography (EMG) across several sessions (between donning/doffing) and within one session (within one donning), regardless of the reason behind the non-stationarity, which can lead to significant deterioration of performance for MPR [16]–[19]. There have been several proposals of adaptive algorithms on this topic [16], [17]. Sensinger *et al.* in [16] proposed a supervised adaptation method, in which new data was reacquired after a period of application for re-training. Alternatively, the unsupervised adaptation proposed by Chen *et al.* [17] provided a promising way to automatically compensate the slow change of EMG signals, however, the experiment in [17] was conducted for only one day without considering the across-session situation. Due to the potential between-session signal non-stationarity, the previous studies suggested the daily re-calibration method. A training strategy combining training data from many days to improve the robustness of the classifier has also been proposed [19], [20]. In either case, the classifier needs to be trained with new data, which can be inconvenient and undesirable for users because of the repetitive training contractions.

In order to reduce the daily re-calibration of classifiers for between-session usage, we previously proposed a method based on domain adaptation (DA), which incorporated the available models from previous sessions into the training process of a new model [21]. Two important

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questions remained. 1) In [21] the DA required a large number of pre-trained models ($N = 9$ in that study), each of which was trained on long data recording from several sessions. Is it possible to reduce the number of pre-trained models, for example, to only reuse one previous model? 2) In [21] we showed that the amount of re-training data required could be very small for the additional session $N + 1$ when N previous models were available. Does the advantage still hold for sessions such as $N + 2$ and $N + 3$, etc., when only the N models are reused? In addition, the non-stationarity of the EMG within one session is another topic which should not be ignored. An ideal MPR system should be capable of compensating such show change of EMG signals. After the short initial training of a new session, the system can be used for a day (about 8 h) without re-training.

To address these issues, we proposed a cascaded adaptation (CA) framework based on linear discriminant analysis (LDA) to reduce the re-calibration time across several sessions and avoid re-calibration within one session. In this framework, on one hand, because of the significant and sudden changes of EMG signals across sessions (for example, due to electrode position shift when donning and doffing), the re-calibration of the classifier in a new session is required. We adopt the DA method which reuses the previous model to reduce the re-training data. One other hand, in order to automatically compensate the slow change of EMG signals during one session application, we use the self-enhancing (SE) method which updates the model parameters according to new data samples and the corresponding recognized labels. In addition, most of the previous studies investigating the adaptive classifiers were based on off-line analysis. However, it has been reported that there only exists weak correlation, if any at all, between offline classification accuracy and online/real-time usability of the system [14], [22]. Therefore, both offline data and online testing are used to comprehensively evaluate the proposed framework in this study. The rest of the paper is organized as the follows. Section II provides the details of the CA framework and the procedures of off-line experiment and online experiment. Section III provides the experiment results. Finally, Sections IV and V describe the result discussion and draw the conclusion, respectively.

II. Methods

A. Formulation of Cascaded Adaptation

1) LDA: LDA is by far the most popular classifier used in MPR because it strikes a good balance of effectiveness and simplicity [17], [23]–[26]. Furthermore, previous studies reported that LDA is more robust than other more sophisticated types of classification techniques [19]. Therefore, here we use the LDA as the classifier.

LDA is based on the Bayesian decision rule and Gaussian assumption [27]. The discriminant function of LDA for a measure x is defined as

$$g_i(x) = \mu_i^T \Sigma^{-1} x - \frac{1}{2}\mu_i^T \Sigma^{-1} \mu_i + \ln p(\omega_i) \tag{1}$$

where μ_i and $p(\omega_i)$ are the mean vector of the training samples and the prior probability of class ω_i, respectively. Σ is the pooled sample covariance matrix which is defined as

$$\Sigma = \sum_{i=1}^{C} \frac{n_i - 1}{N - C} \Sigma_i \tag{2}$$

where C is the number of classes, n_i is the number of the training samples of class ω_i, N is the number of the total training samples and Σ_i is the sample covariance matrix of class ω_i. The class label of the x is determined by choosing the class ω_i which produces the maximum discriminant function value. In sEMG pattern recognition, the numbers of the training samples of all classes generally are the same, and the prior probabilities of all classes are assumed to be equal. Therefore, the term $\ln p(\omega_i)$ in (1) can be omitted and the pooled sample covariance matrix Σ can be calculated as the equally weighted average of the covariance matrices across all classes.

2) Domain Adaptation for LDA: Depending on (1), a LDA model for MPR is determined by the parameters: sample mean vector of each class and pooled sample covariance matrix (the prior probabilities of all classes are assumed to be equal thus omitted). Suppose the pre-stored model from the previous day is represented as (μ_i^o, Σ^o), $i = 1{:}\,C$. The idea of domain adaptation (DA) is that we reuse it and integrate it into the training process of the new day. As described in our previous work [21], the DA for LDA is formulated as

$$\mu_i = r\mu_i^o + (1 - r)\mu_i^n \tag{3}$$

$$\Sigma = r\Sigma^o + (1 - r)\Sigma^n \tag{4}$$

where μ_i^n and Σ^n are the model parameters calculated from the training data of the new day, r is the learning rate which determines the reused degree of the previous model. Comparing with the DA of LDA in [21], we only reuse one previous model in this study to reduce the requirement of pre-trained models, which in [21] the number of pre-trained models was nine.

3) Self-Enhancing for LDA: The self-enhancing (SE) for LDA is an unsupervised adaptation method we previously proposed [17]. It is implemented in the testing stage, where the testing samples and their corresponding recognized labels are fed into the classifier to update the model parameters continuously. Assume that a testing sample (feature) x_t is recognized as class ω_i, then the mean vector of class ω_i and pooled covariance matrix will be updated as the following:

$$\mu_i' = \frac{n_i}{n_i + 1}\mu_i + \frac{1}{n_i + 1}x_t \tag{5}$$

$$\Sigma' = \frac{N - C}{N + 1 - C}\Sigma + \frac{1}{N + 1 - C}S_i. \tag{6}$$

The S_i is defined as:

$$S_i = \frac{n_i}{n_i + 1}(x_t - \mu_i)(x_t - \mu_i)^T \tag{7}$$

where N and n_i are the number of all samples and the number of samples for class ω_i before updating. The derivation of (5) and (6) can be found in [17].

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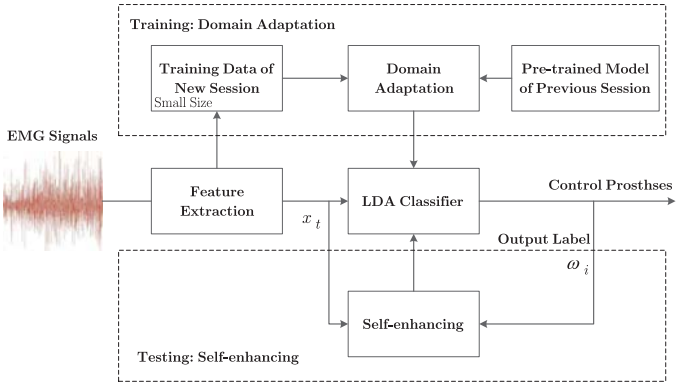

Fig. 1. Scheme diagram of cascaded adaptation framework.

4) Cascaded Adaptation Framework: In the current study, we propose the cascaded adaptation framework to combine the merit of DA and SE. On one hand, the MPR needs to be re-trained because of the relatively significant variation of EMG signals between sessions. The DA is used to reduce the required training samples in the training stage of the new session. On the other hand, the EMG signals can slowly change within one session. The SE is used to automatically accommodate for such change to avoid within-session re-calibration. Fig. 1 demonstrates the scheme diagram of the CA framework, which is attempted to address both between and within session non-stationarity of EMG signals with one single framework.

B. Off-Line Experiment

1) Data Acquisition: Part of the experiment data has been previous reported in [21]. In addition to that dataset, one amputee subjects and three intact-limbed subjects were recruited in the current study. In total, there were eight intact-limbed subjects (all male and right-handed) and three trans-radial amputee subjects (one male and two females) participating in this experiment. The details of the experiment procedure are described as the following.

A commercial wireless biological signal acquisition system, TrignoTM Wireless system (Delsys Inc., USA), was used to record the EMG signals. The skin on the forearm was rubbed lightly with alcohol to provide good electrode-skin contact. For the intact-limbed subjects, four wireless EMG sensors with bipolar configuration were placed on the extensor carpi ulnaris (ECU), flexor carpi radialis (FCR), extensor carpi radialis longus (ECRL) and flexor carpi ulnaris (FCU) of the dominant side, respectively. These four muscles were chosen because they are highly related to human wrist and hand motions. For amputee subjects, because of the difficulty to identify the muscle position, six EMG sensors were placed equidistantly around the circumference of the residual limb. After initial calibration, the sensor positions were marked with a whiteboard marker, so that the electrodes could be placed at the similar positions on subsequent experiment sessions at later days.

The intact-limbed subjects were asked to perform actual hand and wrist motions with the arm where the EMG sensors were placed on. They performed thirteen types of motion in the experiment, including (WF) wrist flexion, (WE) wrist extension, (RD) radial deviation, (UD) ulnar deviation, (P) pronation, (S) supination, (HC) hand close, (HO) hand open, (FP) fine pinch, (KG) key grip, (BG) ball grasp, (CG) cylinder grasp, and (NM) "no motion". The amputee subjects were instructed to produce muscle contractions while attempting a specific movement with their phantom limb. Simultaneously, they performed a mirrored bilateral movement of the intact arm to facilitate the contraction of the affected side. Because of the difficulty to perform BG and CG for the amputee subjects, they only performed the other eleven types of motions.

During the EMG signals recording, the subject was instructed to perform a series of motions at a moderate level of effort. In one trial, each motion was sustained for 5 s, followed by 5 s of rest between motions to avoid fatigue. Each subject performed 20 trials of data capture per session. The time interval between sessions was approximately one day (i.e., one session was performed on a separate day). The entire experiment consisted of 10 days.

The EMG signals were band-pass filtered (20–450 Hz) and sampled at 2 kHz. All data were stored and analyzed offline on a 2.5 GHz Intel Core Quad CPU computer running MATLAB (Mathworks, Inc.).

2) Feature Extraction: In one trial, the contraction data of each motion were segmented into a series of 200 ms windows with 50% overlap. Myoelectric signal features were extracted from each window. The six order autoregressive (AR) coefficients and four time domain features—mean absolute value (MAV), waveform length (WL), zero changes (ZC), slope sign changes (SSC)—were used (called AR+TD) [3], [28]; hence, the dimensionality of the original input feature vector is 40 (4 sensors) for intact-limbed subjects and 60 (six sensors) for amputee subjects.

3) Classification: The classification tasks of intact-limbed subject and amputee subject both included 11 types of movement (i.e., BG and CG of intact-limbed subject were excluded), so that the results of the two groups can be compared. Classification was performed individually for each subject and within each day sequentially (day by day). The purpose of this experiment was to investigate whether the proposed adaptation method can realize fast calibration of MPR, i.e., high classification performance with small re-training data size. The following four classification approaches were compared.

LDA-BL (Baseline LDA): The baseline LDA without adaptation. On the first day, the first 10 trials data were used for training and the other 10 trials data were used for testing. In this way, sufficient training data were used to obtain a reliable LDA model which could be reused on the second day. Also, its high classification performance in the first day would be a benchmark for the following days. From the second day to the last (tenth) day, the size of training data reduced to one trial and the remaining nineteen trials data were used for testing (investigating the adverse impact of small size of recalibration data on non-adaptive method and the potential to compensate such impact through the following adaptive

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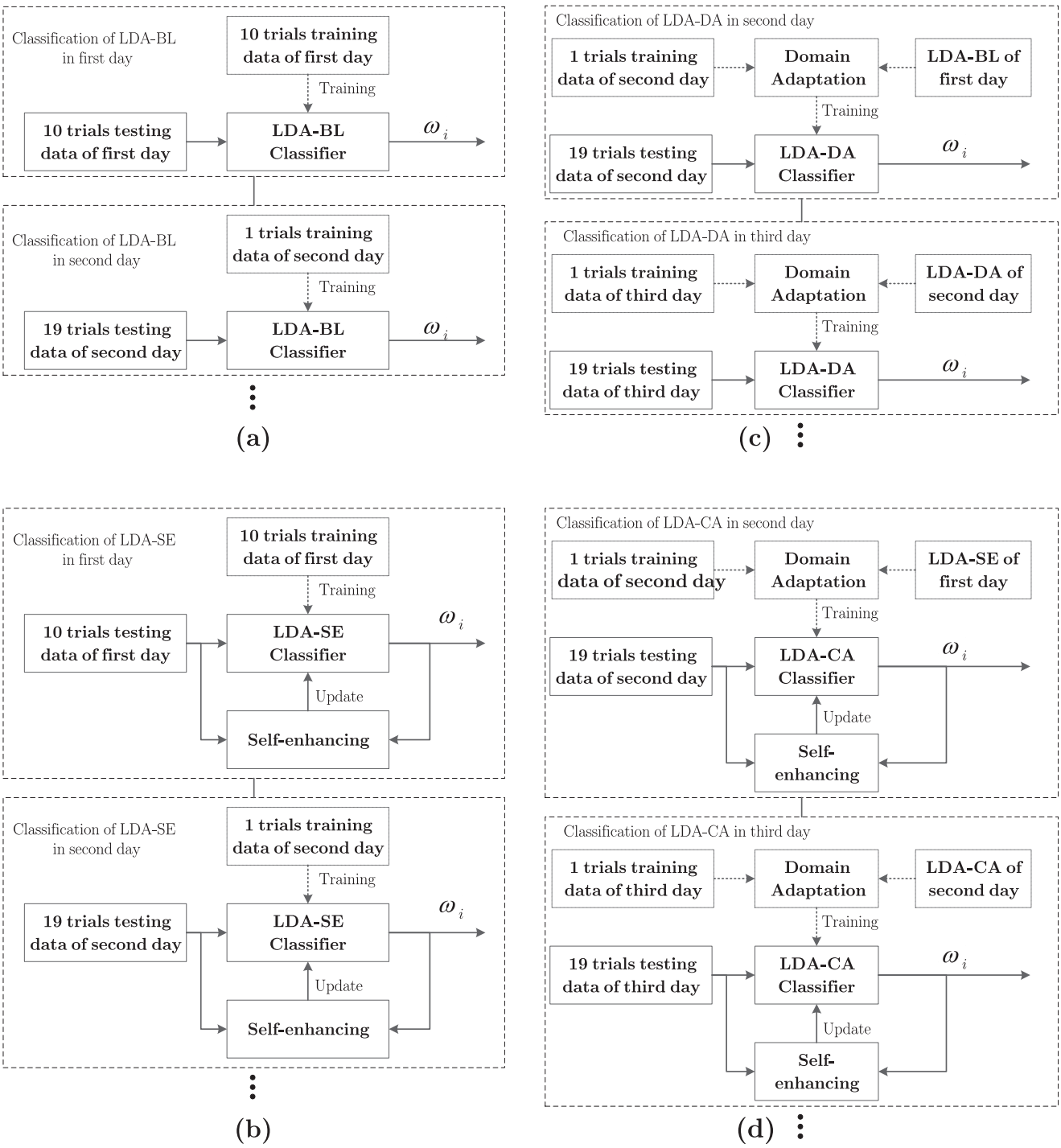

Fig. 2. Data processing flow chart of four classification methods in the off-line experiment. (a) LDA-BL (baseline LDA). (b) LDA-SE (LDA with self-enhancing). (c) LDA-DA (LDA with domain adaptation). (d) LDA-CA (LDA with cascaded adaptation).

approaches). Fig. 2(a) summarizes the data processing flow chart of LDA-BL.

LDA-SE (LDA With Self-Enhancing): The LDA classifier with self-enhancing updating [17]. The proportion of training data to testing data in each day was the same as the description about LDA-BL. In the testing process of each day, the model parameters of LDA-SE were renewed sample by sample, i.e., the updating was performed after recognition of each testing sample. Fig. 2(b) demonstrates the data processing flow chart of LDA-SE.

LDA-DA (LDA With Domain Adaptation): The LDA classifier with domain adaptation. Because the DA algorithm required the storage of at least one previously attained model, the classification of LDA-DA began from the second day. In each day, the sizes of training and testing data were one trial and 19, trials respectively. In the training stage of the second day, the LDA-BL model (not the data) of the first day was reused; at the following days' training stage, the LDA-DA model of the previous day would be reused. The processing flow chart of LDA-DA is illustrated in Fig. 2(c).

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The learning rate r was set to 0.5 according to our previous study [21].

LDA-CA (LDA With Cascaded Adaptation): The LDA classifier with cascaded adaptation. Similar to LDA-DA, the classification of LDA-CA began from the second day, also because the previously attained model was required. The combination of training and testing data was the same as the other three classification methods. In the training stage of the second day, the LDA-SE model of the first day was reused; in the training stage of subsequent days, the LDA-CA model of the previous day was reused. In the testing stage of each day, the model parameters were continuously updated using the SE algorithm. Fig. 2(d) illustrates the data processing procedure of LDA-CA.

4) Performance Evaluation: The widely used metric in MPR, classification rate (CR), was used to evaluate the off-line classification performance. CR is defined as the ratio between the number of correctly classified samples and total number of testing samples

$$\frac{\text{Number of correctly classfication samples}}{\text{Total number of testing samples}} \times 100(\%). \quad (8)$$

Statistical analyses were also performed to investigate whether there exists significant difference among the CR of the four approaches. The multi-factor analysis of variance (ANOVA) contained three main factors (subject, approach and testing day) was used to test the hypothesis. The significance level of all tests was set at 0.05. The details of statistical treatment are provided in Section III-A for clarity of content.

C. Online Experiment

1) Experiment Protocol: This experiment was proposed to investigate whether the system implemented by CA can be used for a day (about 8 h) without recalibration with a fast initial training (about 1 min). Nine intact-limbed subjects (all male and right-handed) participated this online part of the experiment. All subjects have signed the informed consent before the experiment. The experiment procedures conformed to the Declaration of Helsinki. Depending on the off-line experiment results that LDA-SE was slightly better than LDA-DA (not significant, see Section III-A), only three classification methods (LDA-BL, LDA-SE, and LDA-CA) were compared in online experiment to alleviate the burden on the subject.

A commercial biological signal acquisition system, Bluetooth DataLOG (DataLOG W4X8, Biometrics, UK), was used to record the EMG signals. Before experiment the skin on the forearm was rubbed lightly with alcohol to provide good contact for EMG signals acquisition. Four bipolar EMG sensors were placed on the ECU, FCR, ECRL and FCU of the dominant side, respectively. Eleven types of motion were included in the online part of the experiment: (WF) wrist flexion, (WE) wrist extension, (RD) radial deviation, (UD) ulnar deviation, (P) pronation, (S) supination, (HC) hand close, (HO) hand open, (TG) tripod grasp, (KG) key grip, and (NM) "no motion".

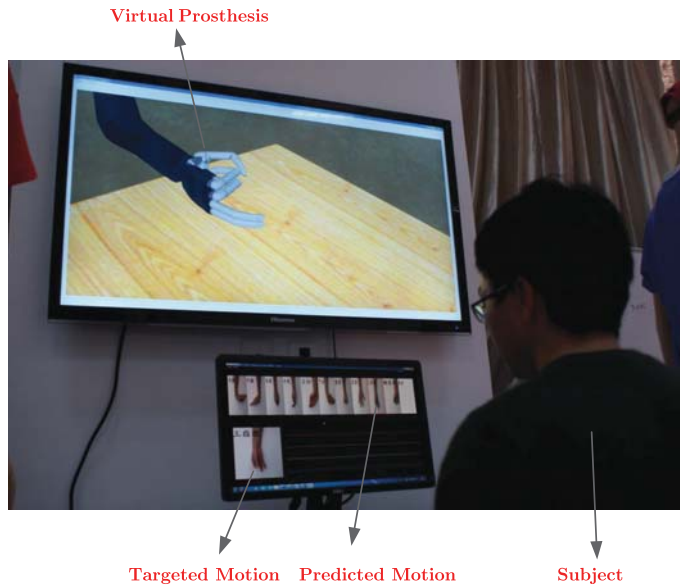

Fig. 3. Illustration of testing stage for online experiment. Subject performs the corresponding motion according to the prompt of target movement. Predicted motion is denoted by red frame to provide real-time feedback for subject.

The experiment was performed with a customized online platform. The platform included EMG acquisition equipment, data processing application running on a PC. The application was written by C++ language. The sampling rate of EMG signals was 1 kHz. The processing window length was 200 ms and the window increment was 100 ms. This meant that the classifier provided a recognized decision per 100 ms during the online testing and the parameters of LDA-SE or LDA-CA were updated per 100 ms. The EMG feature extracted was AR+TD.

The online test was similar to the Motion Test which was originally introduced at the Rehabilitation Institute of Chicago [29], [30]. The definition of a trial in the testing stage was as follows. The PC screen randomly prompted a targeted motion and the subject was instructed to perform the corresponding motion. The prompt of a targeted motion would last for 5 s. During this period the real-time predicted motion would be indicated by a red frame to provide feedback for the subject. Within this 5 s, if the classifier could make 10 correct decisions, a virtual prosthesis was directed to perform the corresponding motion. There was a 5 s break between two targeted motion and each active motion (total 10 types) would appear once in a trial. Fig. 3 illustrates the testing stage in the online experiment.

Each subject participated in a three day experiment. On the first day, 10 trials of training data were collected. In one trial, the subject was instructed to perform the eleven types of motion sequentially according to the prompt on the screen. Each motion was hold for 5 s and there was a 5 s break between two adjacent motions. After the training was finished, two testing sessions were performed. One of testing sessions was performed immediately after the training session, and other second testing session was performed one hour after the first testing session. In each testing session, eight trials of evaluation was performed for each of the two classification

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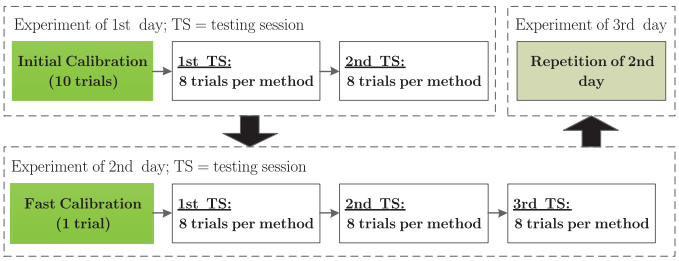

Fig. 4. Diagram illustrating the procedure of the three day online experiment for each subject.

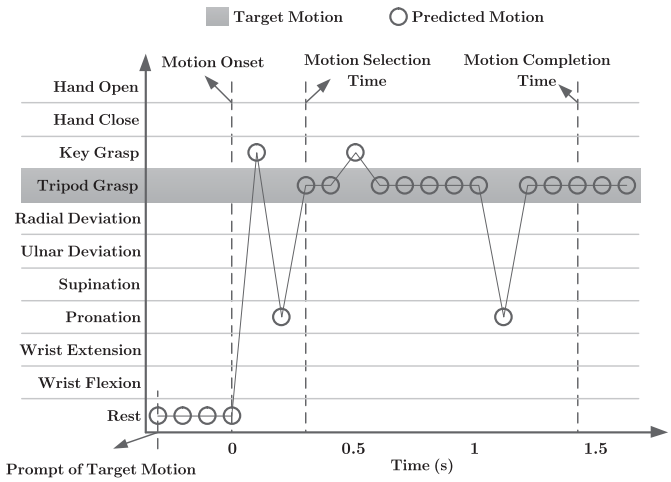

Fig. 5. Demonstration of performance measure of online experiment. Motion onset, motion selection time and motion completion time is showed here according to the classifier prediction output.

approaches (LDA-BL and LDA-SE). In a testing trial a fixed classification approach was used. The classification approaches in different trials were randomly provided. On the second day, only one trial of training data were collected, and three testing sessions were performed. Also, each of the three classification approaches (LDA-BL, LDA-SE, and LDA-CA) was assessed by eight trials of testing in each testing session. The first testing session and second testing session was about 2 h apart, including resting periods and a lunch. The interval between the second session and third session is about one hour. The EMG electrodes were not removed between testing sessions and the subjects could freely have a rest. The experiment procedure of the third day is the same as the second day. Consequently, the experiment of the first day lasted about 4 h and the experiment of the second day and third day lasted about 8 h. The entire experiment procedure is illustrated in Fig. 4.

2) Performance Evaluation: We used four metrics to evaluate the online classification performance, three of them were previously reported in [30].

Motion Selection Time (MST): The MST denotes the time from the motion onset when a certain targeted motion appears to the time when the first correct decision is produced (see Fig. 5). The motion onset is defined as the time of the last decision of rest after the prompt of a targeted motion. MST reflects the responsiveness of the classification system. If the targeted motion is not correctly selected within the 5 s (duration of the prompt), the MCT is not counted.

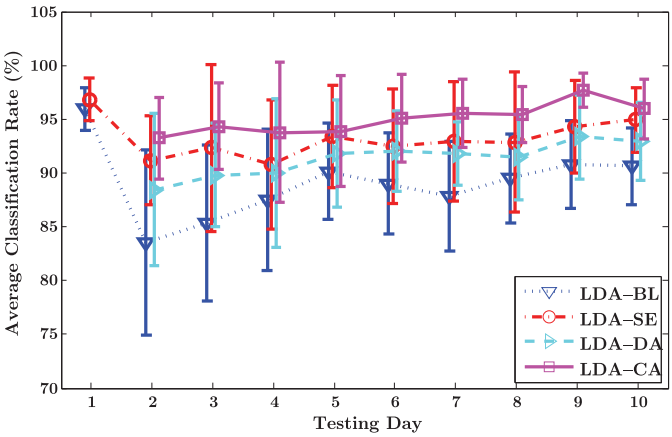

Fig. 6. Comparison of average classification rates among four classification methods in each day for intact-limbed subjects. Error bars represent ±std.

Motion Completion Time (MCT): The MCT denotes the time from the motion onset to the completion of the targeted motion. The time of motion completion is defined as the time of tenth correctly predicted decision after the prompt of a targeted motion (see Fig. 5). MCT reflects the stability of the classification.

Motion Completion Rate (MCR): If not less than 10 correct decisions are obtained within the 5 s of the prompt of a targeted motion, the motion was completed. The MCR is defined as the ratio of the number of completed motions to the total number of motions. If the motion is not completed, the MCT is not counted.

Real-Time Accuracy (RTA): RTA is defined as the percentage of the correctly classified samples from the motion onset to the end of the prompt of a targeted motion.

Statistical analyses were also performed to investigate whether there exists significant difference among each of the four metrics of the three methods. The details of statistical treatment are provided in Section III-B for clarity of content.

III. Results

A. Off-Line Experiment

1) Intact-Limbed Subjects: The average CRs of four classification methods across eight intact-limbed subjects in each day are illustrated in Fig. 6. The classification results of LDA-BL and LDA-SE from first day to tenth day are showed while the results of LDA-DA and LDA-CA from second day to tenth day are showed. Comparing with the CR of the first day, the CR of LDA-BL evidently reduced in the second day when limited number of training samples was available (>10%). When the size of training data was small (from second day to tenth day), the three classification methods with adaptation all were consistently better than the LDA-BL. LDA-CA achieved the best classification performance as the average CR in each day was higher than 93% even when only one trial data were used for training. In addition, we observe a trend that the average CRs of four classification methods increase from the second day to the tenth day, especially for LDA-BL. This possibly originates from the fact that there exists the learning effect of subjects when their experiment experience increased [31], [32].

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TABLE I
TUKEY COMPARISON OF CLASSIFICATION METHODS FOR CLASSIFICATION RATE

	Classification Methods			
	LDA-BL	LDA-SE	LDA-DA	LDA-CA
	Intact-limbed Subjects			
Mean±Std	88.24±5.80	92.80±5.27	91.29±4.82	94.99±3.97
Groups	C	B	B	A
	Amputee Subjects			
Mean±Std	79.94±6.97	85.05±5.90	83.81±7.47	88.67±4.42
Groups	C	B	B	A

Because the focus of this study is on the classification performance when the size of training data was small, the ANOVA was performed on the data from the second day to the tenth day. First, the full model which included three main factors (subject, method and testing day) and all paired interaction factors showed that subject had interaction with other two factors. In order to reduce the complexity of analysis, reduced ANOVA model which included subject (random factor, 8 level), method (fixed factor, 4 level) and testing day (fixed factor, 9 level) as main factors and one interaction factor (method versus testing day) was performed. Results showed that there was no significant interaction between method and testing day ($p = 0.997$). Therefore, the ANOVA was reduced to the linear model which only contained the three main factors. Results showed that all three main factors had significant effect on the CR ($p < 0.01$). And then, the Tukey comparison with correction was performed for classification methods to investigate the paired difference. The results are summarized in Table I, where if the group memberships of two methods (see second row) are the same, there is not significant difference between them. Therefore, the result of Tukey comparison is that LDA-CA > LDA-SE = LDA-DA > LDA-BL. Furthermore, we find that not only the average CR of LDA-CA is the highest but also the standard deviation (STD) of LDA-CA is the smallest among the four methods, indicating the superior consistency of LDA-CA under different experiment conditions (subject and testing day).

2) Amputee Subjects: The average CRs of four classification methods across three amptuee subjects in each day are depicted in Fig. 7. In general, the classification performance of amputee subjects was inferior to that of intact-limbed subjects. Nonetheless, the trends of comparison among the four methods were similar for the two subject groups. When the number of training samples was limited, the three adaptation methods all outperformed the baseline method and the LDA-CA was the best. The average CR of LDA-CA in each day was higher than 85% when only 1 trial data were used for training. Also, the CR increases from the former day to the latter day for LDA-BL.

Similar to the statistical test of intact-limbed subjects, the ANOVA model including three main factors (subject, method and testing day) and one interaction factor (method versus testing day) showed that there was no significant interaction between method and testing day ($p = 0.944$). And then, the results of linear ANOVA showed that all three main factors had significant effect on the CR ($p < 0.01$). Tukey comparison of classification methods are summarized in Table I. The results of statistical test are the same as those of intact-limbed subjects.

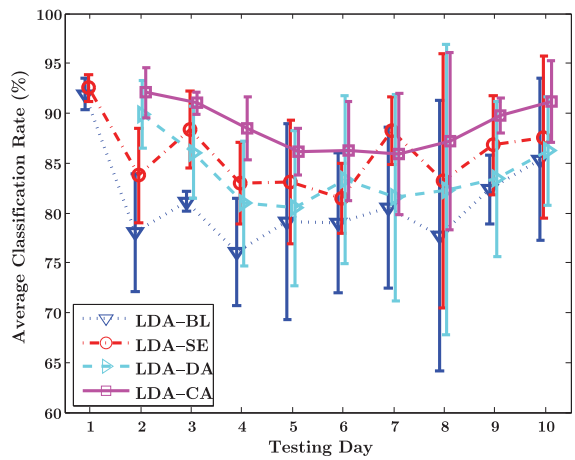

Fig. 7. Comparison of average classification rates among four methods in each day for amputee subjects. Error bars represent ±std.

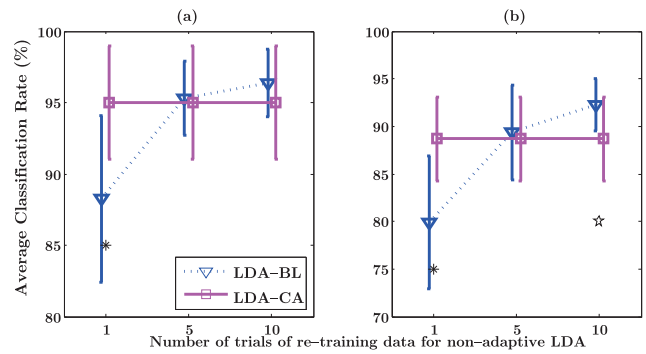

Fig. 8. Comparison of average classification rates between LDA-CA with LDA-BL when re-training data of LDA-BL increased. (a) Intact-limbed subjects; (b) amputee subjects. Error bars represent ±std. "$*$" represents LDA-CA significantly better than LDA-BL while "$\star$" represents the opposite.

3) Adaptive Algorithms Versus More Re-Training Data: In order to further demonstrate the potential benefits and re-training time savings of LDA-CA, we compared the LDA-CA with the LDA-BL when the re-training data of LDA-BL increased from one trial to five trials and 10 trials. The results of intact-limbed subjects and amputee subjects are illustrated in Fig. 8(a) and (b), respectively. The classification rates were averaged across nine testing days. We see that the performance of LDA-CA was equal to that of LDA-BL which included five trials of re-training data while was a little inferior to the full retraining LDA-BL (10 trials). This demonstrates that the proposed adaptive algorithm can be approximately re-calibrated five times faster than the non-adaptive baseline method.

B. Online Experiment

The average MST, MCT, MCRs, and RTAs of LDA-BL, LDA-SE, and LDA-CA across nine intact-limbed subjects are demonstrated in Fig. 9. The x-axis label "D1-S1" denotes the first testing session in the first day and the other x-axis labels

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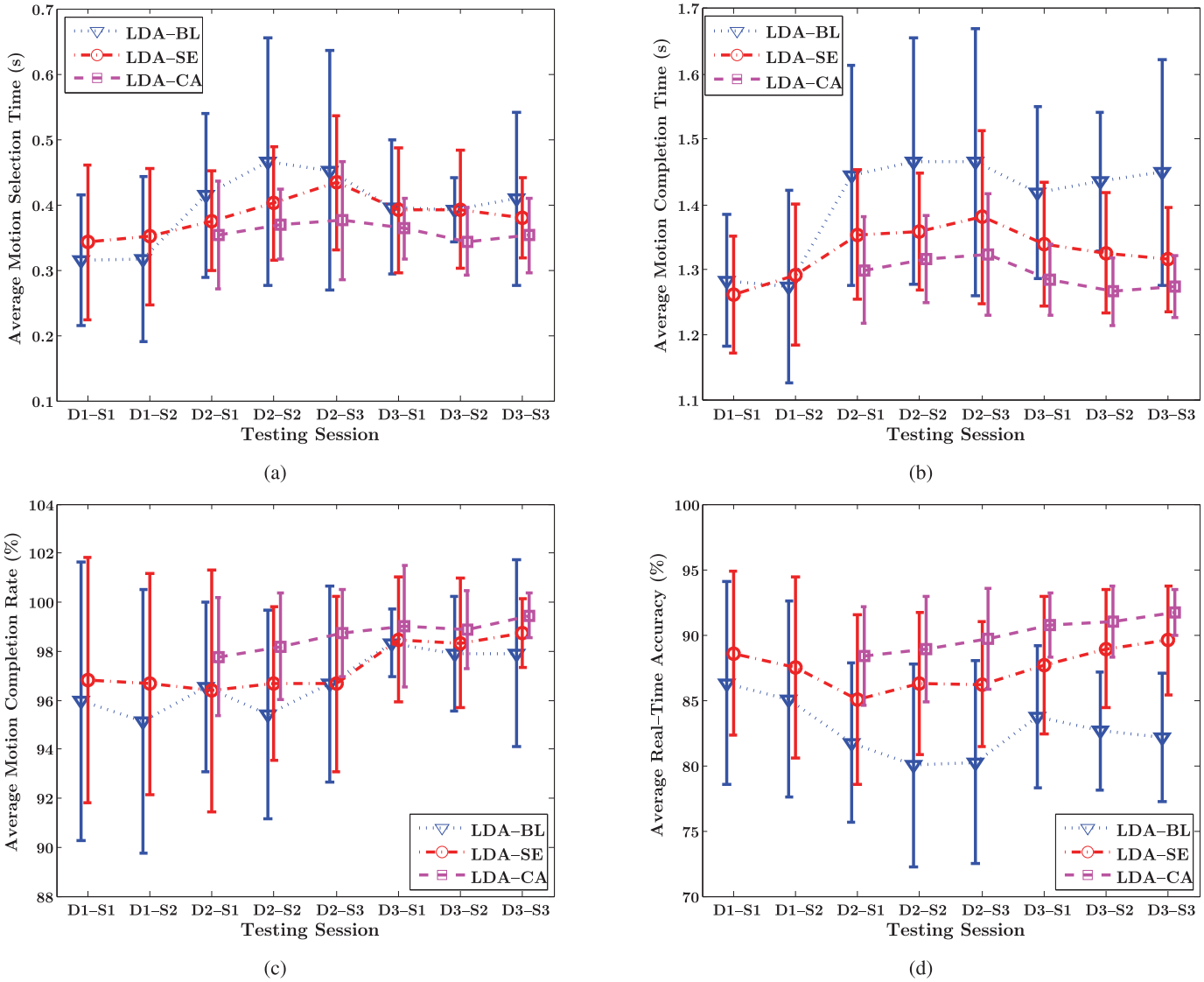

Fig. 9. Comparison of four online performance metrics among three methods in each testing session. (a) Motion Selection Time. (b) Motion Completion Time. (c) Motion Completion Rate. (d) Real-Time Accuracy. Error bars represent $\pm std$.

follow the same pattern. The LDA-CA outperforms the other two methods for all four performance metrics, not only with better average value but also with a smaller STD. The LDA-SE outperforms LDA-BL in MCT and RTA. The MCRs for three methods are all fairly high, with average > 95% in each testing session.

Furthermore, the statistical analysis was performed on the data from the second day and third day for each online metric. The ANOVA model which included subject (random, 9 level), method (fixed, 3 level), testing day (fixed, 2 level) as main factors and one interaction factor (method versus testing day) showed that there was no significant interaction between method and testing day (MST: $p = 0.558$; MCT: $p = 0.906$; MCR: $p = 0.464$; RTA: $p = 0.843$). Therefore, the ANOVA was reduced to linear model. Except that the testing day had not significant effect on MST ($p = 0.07$), all the other tests showed significance ($p < 0.05$). Then, the Tukey comparison of classification methods was performed on the four performance metrics. The results are summarized in Table II.

TABLE II
TUKEY COMPARISON OF CLASSIFICATION METHODS FOR FOUR ONLINE PERFORMANCE METRICS

Performance Metric		LDA-BL	LDA-SE	LDA-CA
MST (s)	Mean±Std	0.42±0.14	0.40±0.08	0.36±0.06
	Groups	A	A&B	B
MCT (s)	Mean±Std	1.45±0.16	1.35±0.10	1.29±0.07
	Groups	A	B	C
MCR (%)	Mean±Std	97.13±3.36	97.55±3.23	98.68±1.95
	Groups	B	B	A
RTA (%)	Mean±Std	81.76±6.05	87.29±5.17	90.07±3.28
	Groups	C	B	A

MST = Motion Selection Time; MCT = Motion Completion Time; MCR = Motion Completion Rate; RTA = Real-Time Accuracy.

As in row of MST, the membership label "A&B" represents that LDA-SE belongs to not only Group A but also Group B, therefore, LDA-SE is not significantly different from LDA-BL and LDA-CA but the MST of LDA-CA is significantly lower

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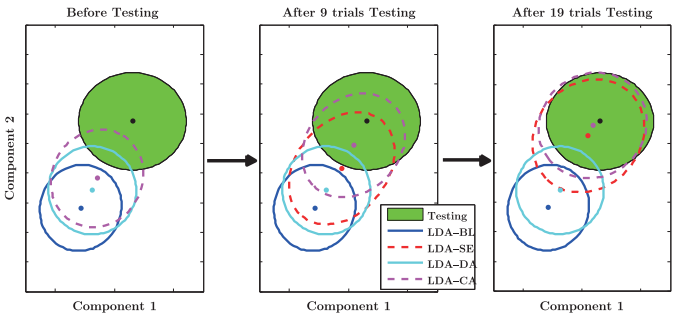

Fig. 10. Illustration of coincidence between model parameters of each classification method and testing data. Data from a certain testing day of a representative subject. Motion type showed here is pronation.

than that of LDA-BL. The LDA-CA significantly outperforms LDA-BL in all four metrics and significantly outperforms LDA-SE in MCT, MCR and RTA. The LDA-SE significantly outperforms LDA-BL in MCT and RTA. In addition, the STD of LDA-CA is the smallest for all four performance metrics, demonstrating its robustness.

IV. Discussion

A. Effectiveness of Cascaded Adaptation

Results of both off-line and online experiments demonstrate that the adaptive LDA methods outperform the baseline LDA when the available training samples are limited and the LDA-CA obtains the best performance. LDA-CA provides a potential tool to realize high performance of myoelectric control using only one trial training data as shown in Figs. 6, 7, and 9. In particular, the results in Fig. 9 show that the myoelectric control system with LDA-CA is able to be used for 8 h without re-training when initial training time is about 1 min. The LDA-CA first uses the DA algorithm to improve the usability of the re-trained classifier in a new day and then use SE algorithm to compensate the slow change of EMG signals. In order to visualizing this process, we use the data in a certain testing day (after the second day) of a representative subject for analysis. We know that there are one trial training data and 19 trials testing data in this day. The following parameters can be computed from the testing data: $(\mu_i^{test}, \Sigma^{test})$, $i = 1{:}\,C$, where μ_i^{test} is mean vector of class ω_i and Σ^{test} is pooled covariance matrix. Also, we have the model parameters of each classification method: $(\mu_i^{BL}, \Sigma^{BL})$, $(\mu_i^{SE}, \Sigma^{SE})$, $(\mu_i^{DA}, \Sigma^{DA})$, and $(\mu_i^{CA}, \Sigma^{CA})$. These parameters were projected into 2D plane through the first two leading eigenvectors of Σ^{test} for visualization, as shown in Fig. 10. The left sub-figure represents that the initial training is just finished; the middle sub-figure represents that nine trials data have been tested; the right sub-figure denotes that all 19 trials data have been tested. Note that before testing the model parameters of LDA-BL and LDA-SE are the same, therefore the LDA-SE model is not displayed in the left sub-figure. LDA-BL and LDA-DA are static classifiers and their parameters do not change during the testing, while the parameters of LDA-SE and LDA-DA approach to those of testing data as the testing runs.

In addition, LDA-SE is not just abe to compensate the slow change of EMG signals. The superiority of LDA-SE compared with LDA-BL when training data set is small illustrates that LDA-SE has the role like LDA-DA to reduce the training time. The SE method has also been used to electroencephalography (EEG) classification when training data reduced [33]. Therefore, the LDA-CA is not simply a linear combination of DA and SE methods, i.e., simply a function plus of "reducing retraining data" and "compensating variation of EMG". There exists coaction between DA and SE in the framework of LDA-CA. On one hand, the DA method helps to produce a reasonable initial model for performing SE. Otherwise, if the initial model is poor, the classifier labels the testing samples with too much error decisions and it can not effectively follows the change of EMG patterns. On other hand, the LDA-DA here is not good enough with a small number of training data (see Figs. 6 and 7), the SE method provides a way to augment the training data size with recognized testing samples, thereby reducing the requirement of pre-collecting of training data [33].

B. Comparison Between Off-Line Experiment and Online Experiment

Previously studies have reported that the correlation between offline classification accuracy and online usability is not necessarily evident [14], [22]. The important difference between offline and online performance of classification algorithms is that the subject can receive real-time feedback when performing online experiment. The real-time feedback is able to help the subject to modulate his/her muscle contraction pattern to improve the online control, as demonstrated in [34], where the real-time accuracy is about 85% while the system usability is fairly good. This is further validated in our experiment. As showed in Fig. 9, although the RTA of LDA-BL is usually lower than 85%, the MCR of LDA-BL can be higher than 95%. The RTA of the third day is lower than that of the first day, but the MCR is higher. This can be attributed to the fact that the subjects were able to better modulate muscle contraction after two days' experiment. In addition, the decrease of RTA of LDA-BL ($< 5\%$, see Fig. 9) is not as much as the decrease of CR ($> 10\%$, see Figs. 6 and 7) when the training data reduce from the first day to the second day. The RTA of LDA-BL does not decrease significantly during the testing within one day, different from the results in [17]. The statistical test also shows that the testing session has not significant effect on the all four online performance metrics. Furthermore, thanks to the real-time feedback, the LDA-SE and LDA-CA improve their performances for some subjects among the testing sessions within one day. The improvement likely originates from the co-adaptive effect of the classifier and the subject. This co-adaptation between machine and subject has also been reported in [35], [36] and can further supports the benefits of LDA-CA. As we see in Fig. 8, the LDA-CA is a little inferior to the full retraining baseline method in off-line data. However, in online experiment the LDA-CA appears to be equal to or even better than the full training in first day (see Fig. 9). In summary, the difference

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between the offline experiment and online experiment emphasizes the necessity of real-time evaluation for myoelectric control.

C. Learning Effect of Subjects

Both offline and online experiments showed that the system performance increased with the subject's experience with myoelectric control time increased. The statistical test further validates this conclusion: the testing day has significant effect on the experiment results. These results demonstrate that there exists the learning effect of subjects as the experiment proceeds. Such learning effect has also been investigated in [31], [32], [37]. Therefore, it is expected that the performance of each classification method can further improve as the number of experiment day increases. Nevertheless, the proposed CA framework provides a promising way to fulfill the fast calibration of myoelectric control. Beginning from the second day, it has been able to obtain excellent performance with small training data. Furthermore, the proposed LDA-CA is embedded with the co-adaptation between the machine and user. Such co-adaptation provides a possible way to accelerate the subject training, comparing with the open-loop training method described in [31], [32], [37].

V. Conclusion

In this paper, a CA framework is proposed to compensate for the fast change of EMG signals during across-session usage (across several donning/doffing) and slow change of EMG signals during within-session usage (within one donning/doffing), to enable fast calibration for myoelectric control. The proposed CA is able to improve the classifier capacity by re-using the previous model in the training stage and continuously updating the model parameters in the testing stage. The offline experiment which includes 11 subjects (eight intact-limbed subjects, three amputees) and the online experiment which included nine intact-limbed subjects both demonstrate the effectiveness of the proposed method. The LDA-CA can be used for a day (about 8 h) without re-training with a short (about 1 min) initial training, thereby significantly reducing the training requirement for MPR.

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